Experimental Investigation of Airfoil Aerodynamics and Flow Control Concepts: Mono and Biplane Configurations

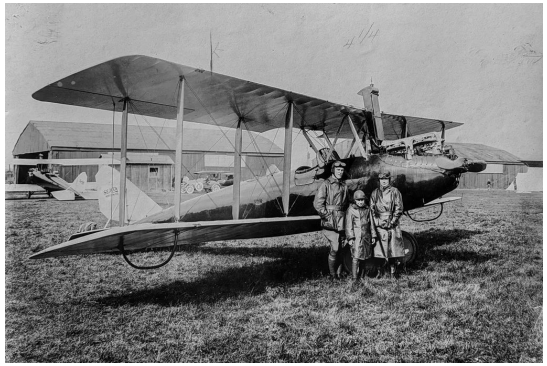
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Motivation

- Biplane interference theory dates back to Munk (1922) and Prandtl (1924), biplanes were abandoned before modern CFD and testing existed
- There is almost no modern experimental data on how wings interact at Low Re
- Advantages of Biplanes: low speed, high lift, short wingspans, short-field use



Oriole, 1919, Cradle of Aviation Museum



TVS-2DTS

Background: Biplane Theory

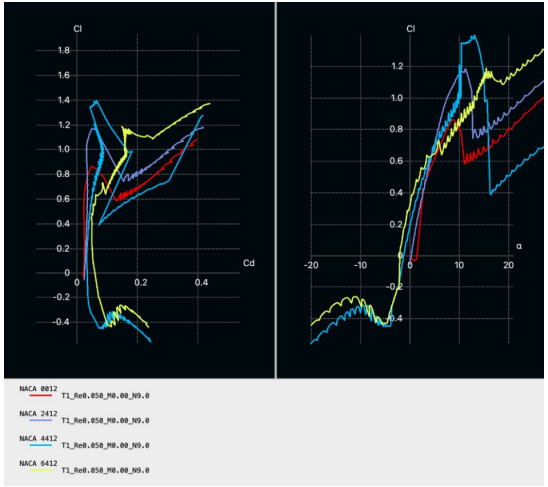
- Biplane geometry is defined by gap (vertical spacing), stagger (horizontal offset), and decalage (relative angle between wings)
- The Kutta condition is satisfied at both trailing edges by mapping to an equivalent tandem arrangement and solving for the circulation
- Drag Mechanisms:
  - Sectional Interference: pressure fields overlap across the gap
  - Induced-Drag Interference: trailing (tip) vortices interact; finite span only

β=incidence, β_0= camber angle, B<1 → biplane lift-loss factor

$$C_L = 2\pi \sin\beta\, B + 2\pi \sin\beta_0 \sqrt{B}$$

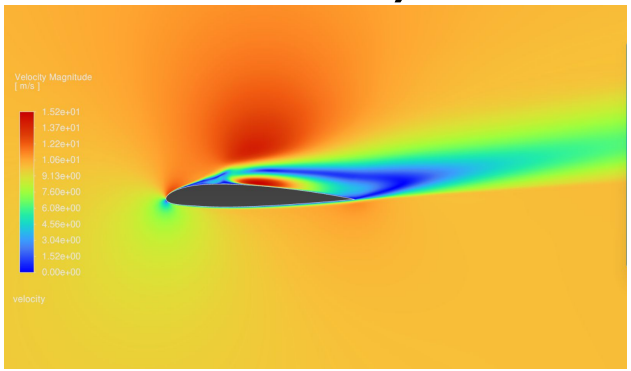
Initial Analysis: Simulation

XFLR5 (Panel + BL)



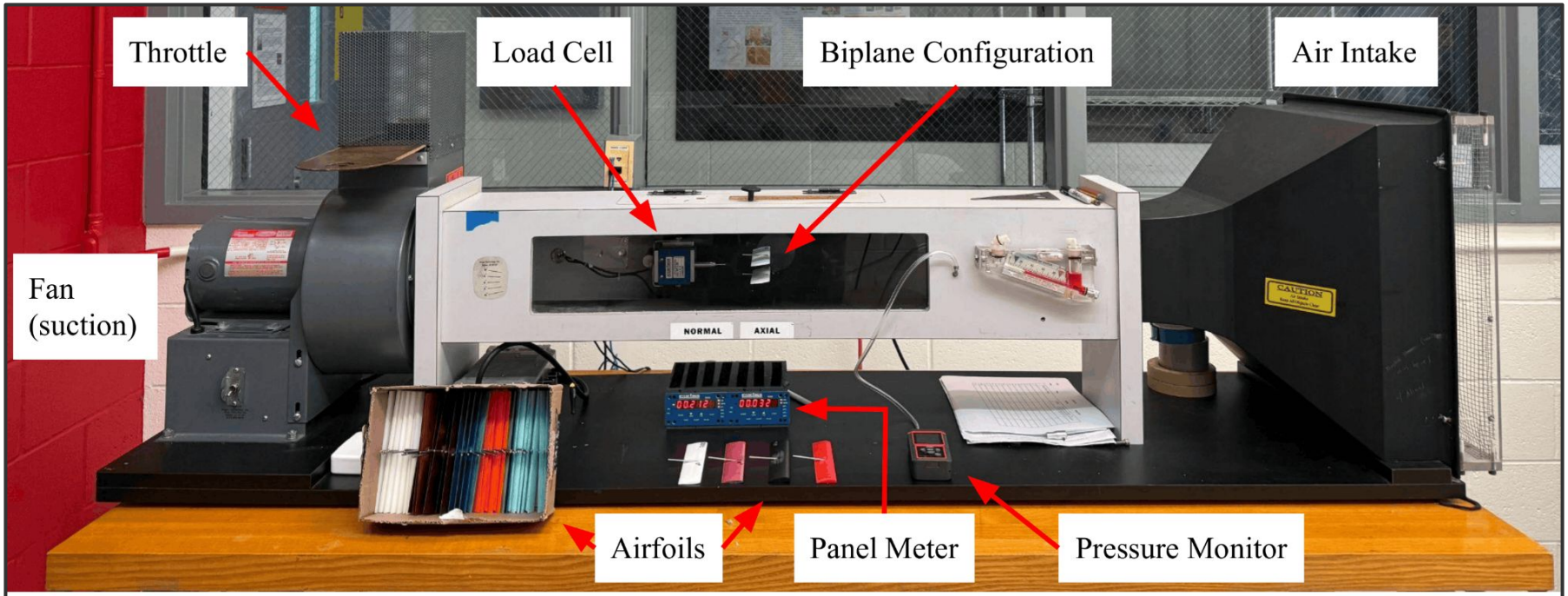
Preliminary screening of 16 airfoils. Trials focused on the effects of each NACA digit

Ansys Fluent (2D RANS)



Simulated the top 6 from XFLR5 (streamlines, velocity contours)

Experimental Procedure



Re = 20,100 (about 10 m/s), 30 mm chord, AR 5, 30 mm gap, no stagger

Wind Tunnel Test Matrices

Test Matrix 1: Airfoil Shape

	0012 (A)	2212 (B)	2412 (C)	2612 (D)
0012 (A)	AA	BA	CA	DA
2212 (B)	AB	BB	CB	DB
2412 (C)	AC	BC	CC	DC
2612 (D)	AD	BD	CD	DD

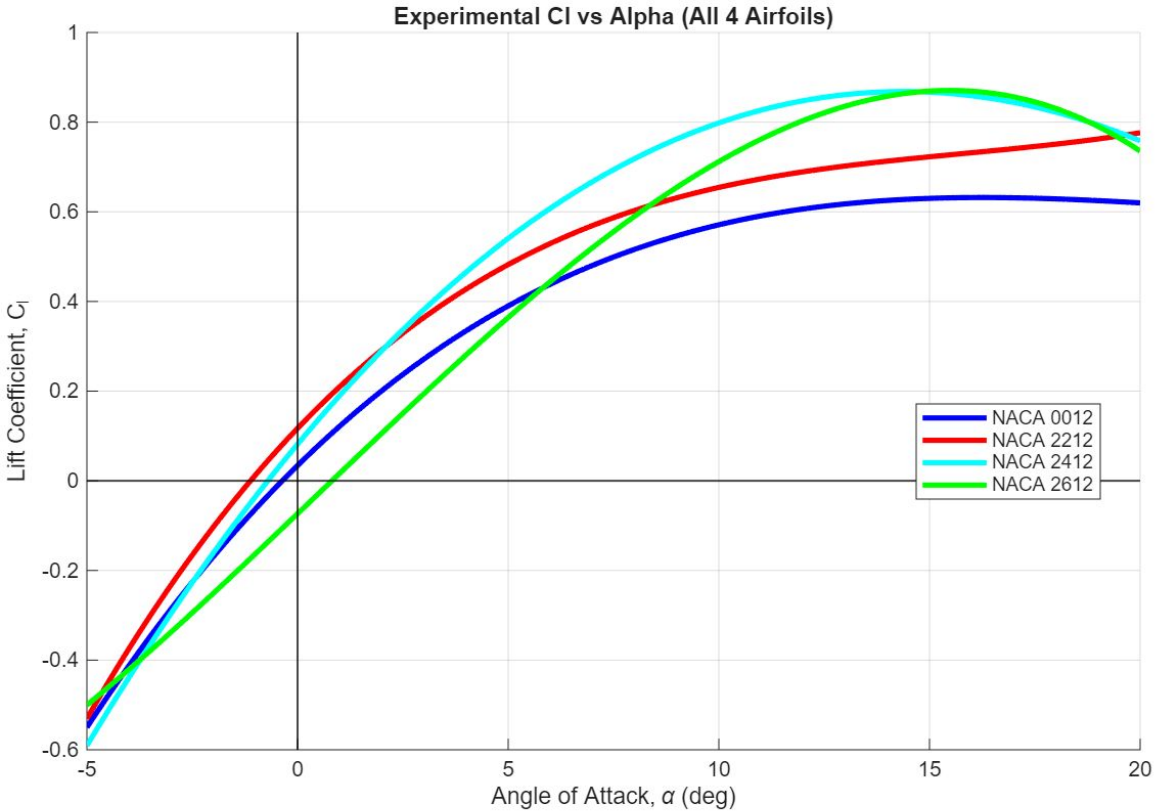
Test Matrix 2: Airfoil Angles

Top Wing α

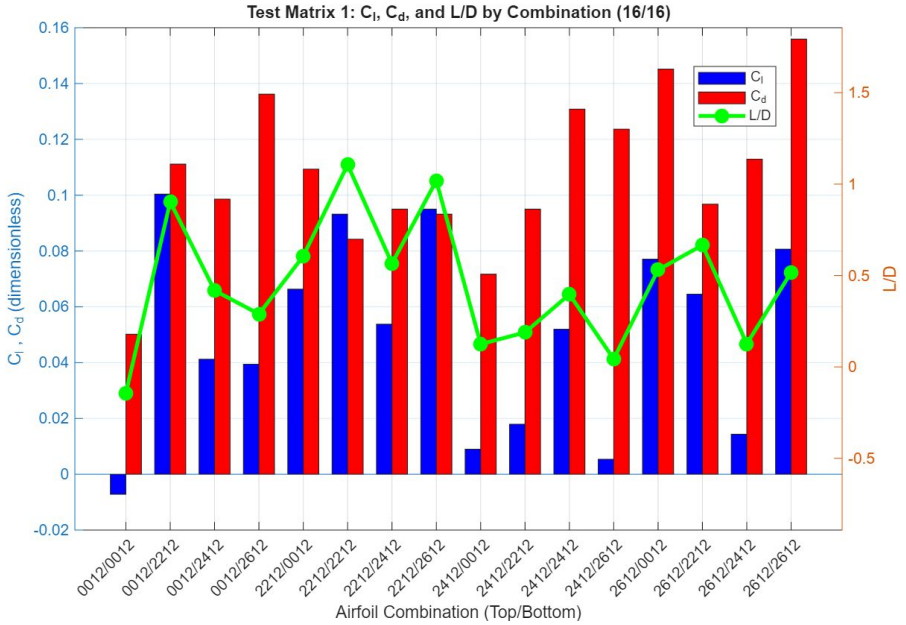
Bottom Wing α	-5°	0°	5°	10°	15°	20°
-5°	-5°,-5°	0°,-5°	5°,-5°	10°,-5°	15°,-5°	20°,-5°
0°	-5°,0°	0°,0°	5°,0°	10°,0°	15°,0°	20°,0°
5°	-5°,5°	0°,5°	5°,5°	10°,5°	15°,5°	20°,5°
10°	-5°,10°	0°,10°	5°,10°	10°,10°	15°,10°	20°,10°
15°	-5°,15°	0°,15°	5°,15°	10°,15°	15°,15°	20°,15°
20°	-5°,20°	0°,20°	5°,20°	10°,20°	15°,20°	20°,20°

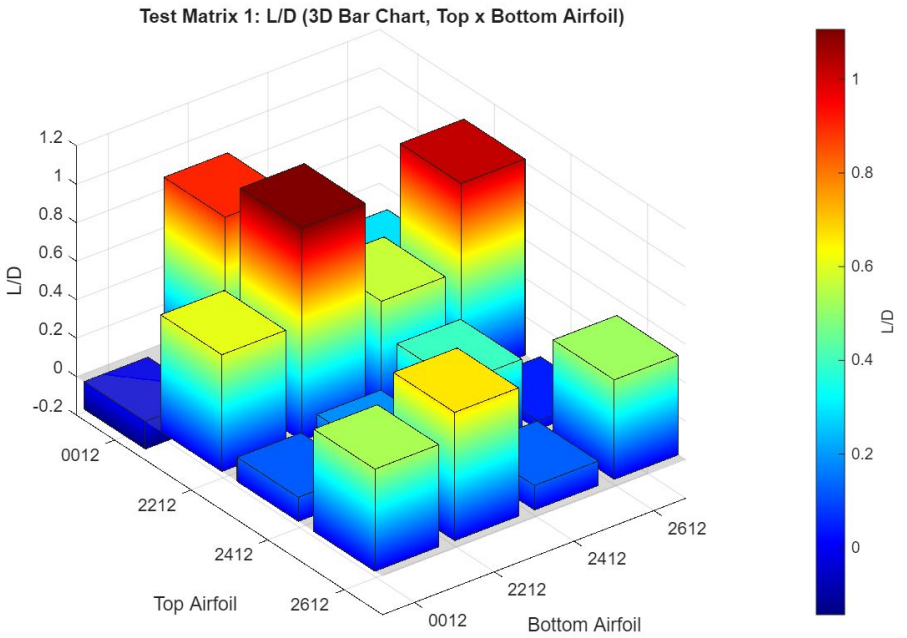
Monoplane Results

- 2212 highest C_L near 0°
- 2412 and 2612 reach the highest C_L (≈0.87 near 15°)
- 2212 shows no stall through 20°; laminar separation bubble



Biplane Results: Airfoil Shape Combinations





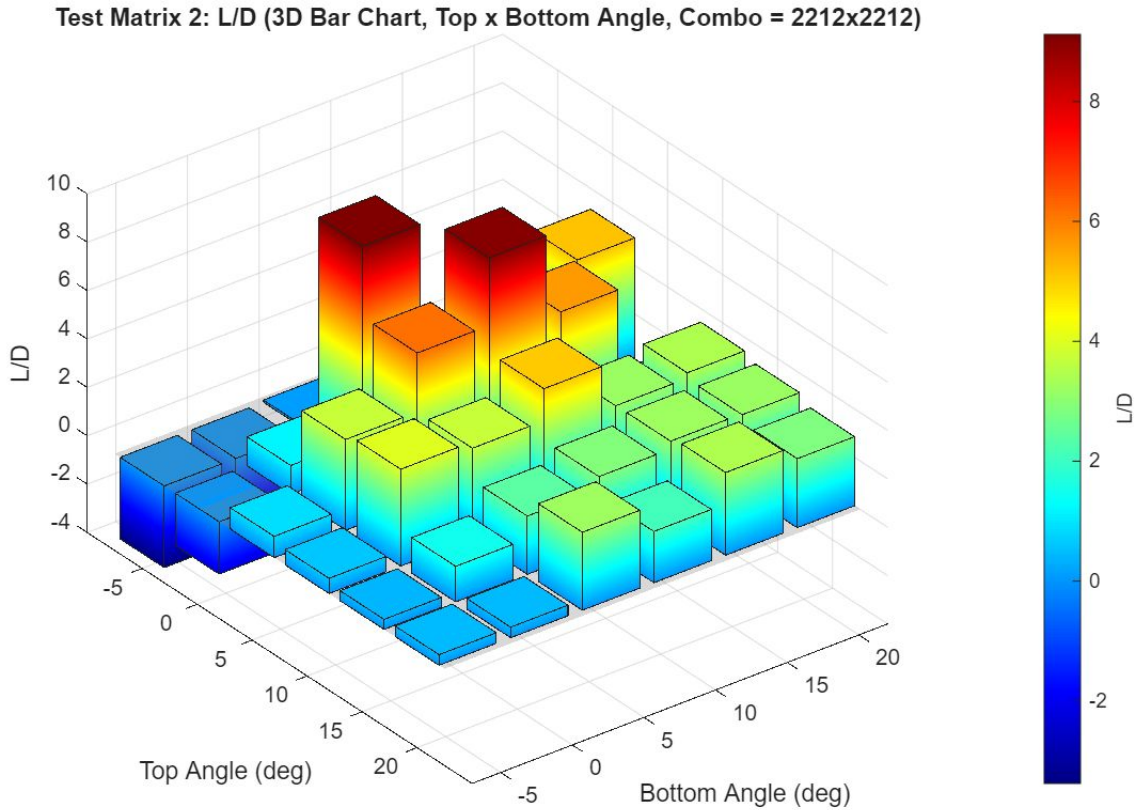
Biplane Results: Decalage Optimization

$$C_L = 2\pi \sin\beta\, B + 2\pi \sin\beta_0 \sqrt{B}$$

- Lift rises with mean incidence
- Drag is ignored, so the equation cannot rank L/D

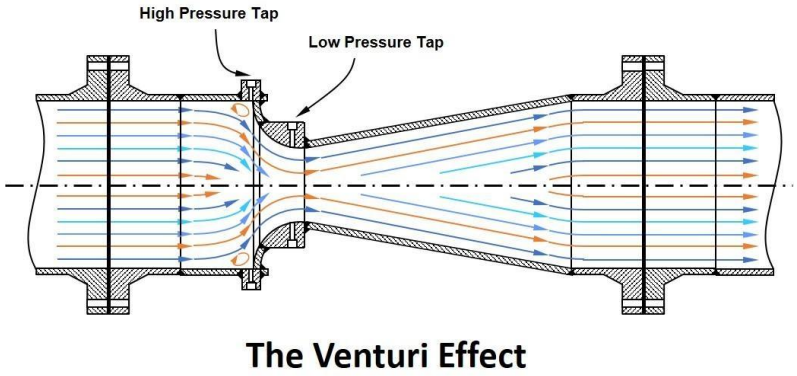
0° × 5°: L/D = 9.12
5° × 10°: L/D = 8.98

Notation used:
Top x Bottom



Discussion: Why the Best Configurations Win

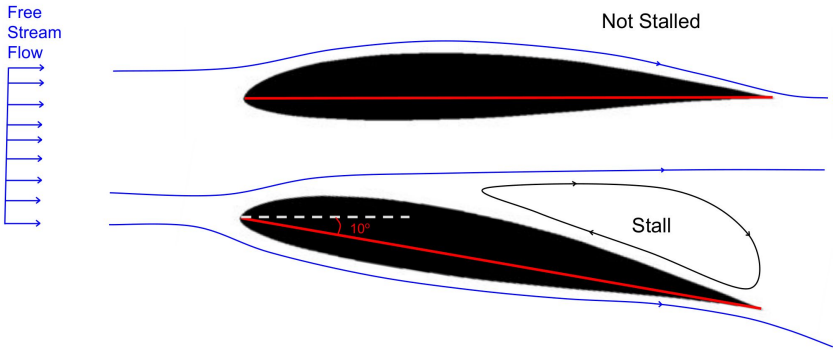
Concentrated Venturi Effect



The Venturi Effect

Venturi Operating Principle, Wyatt Engineering
Inferred from force data, not measured

Stall-Order Asymmetry



Conclusions & Future Work

- 9.12 vs 4.21 (biplane vs monoplane)
- NACA 2212 x 2212 was the most efficient pairing (0° x 5°)
- Biplane performance depends on wing-to-wing interaction
- Applications include more efficient low-speed/low Re aircraft; potential variable geometry designs using a retractable second wing.
- Key limitations: assumed turbulence intensity (0.5%) and Oswald efficiency factor (e=0.8) in the finite-span correction; wall proximity and tip leakage; the tunnel is near-2D, so induced-drag interference was not measured.
- Future work: optimize stagger and gap alongside decalage and airfoil shape; extend to supersonic speeds and the theoretical Busemann biplane.